

Ethical & Responsible AI

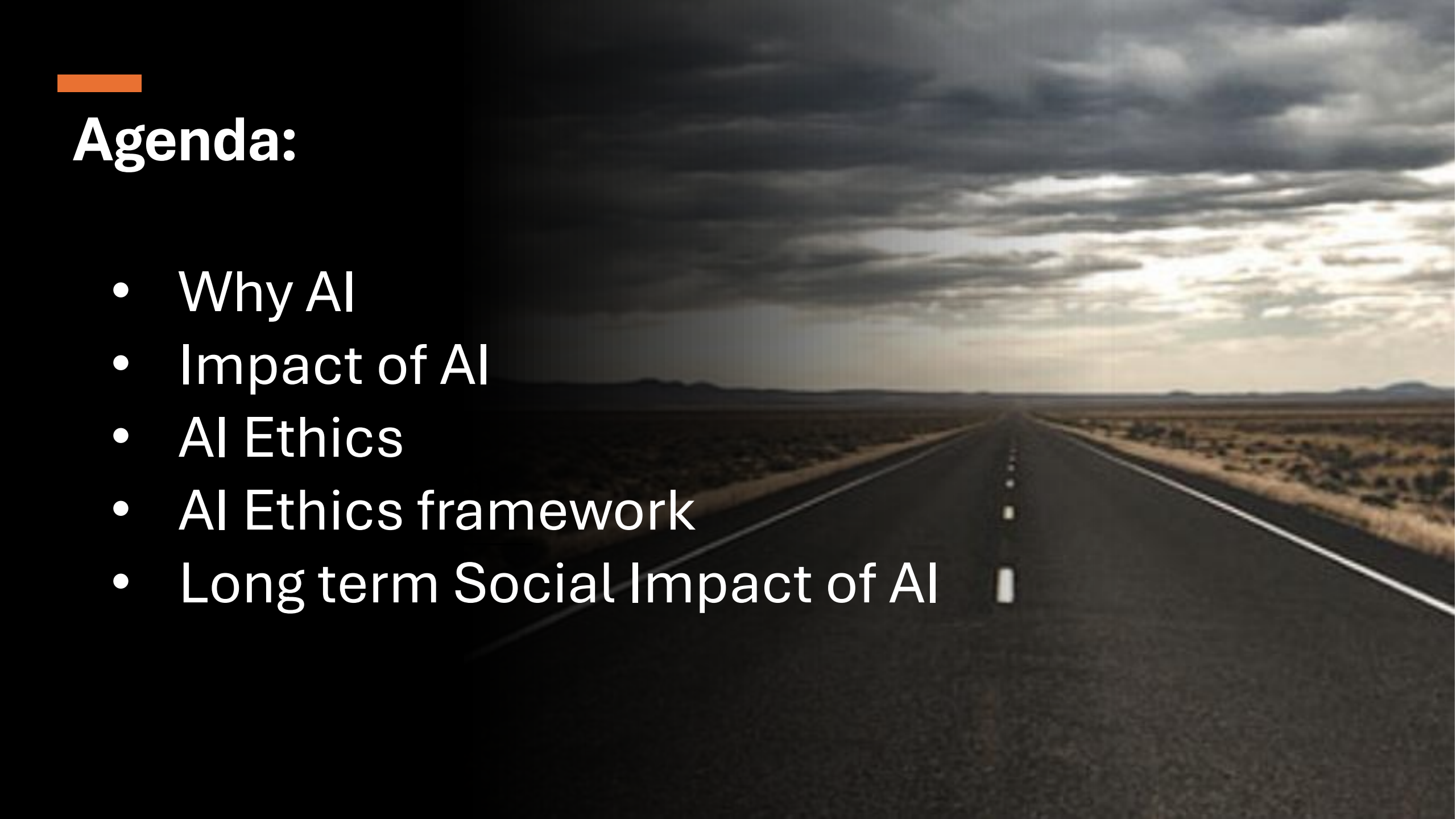
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UNIVERSITY OF RUHUNA

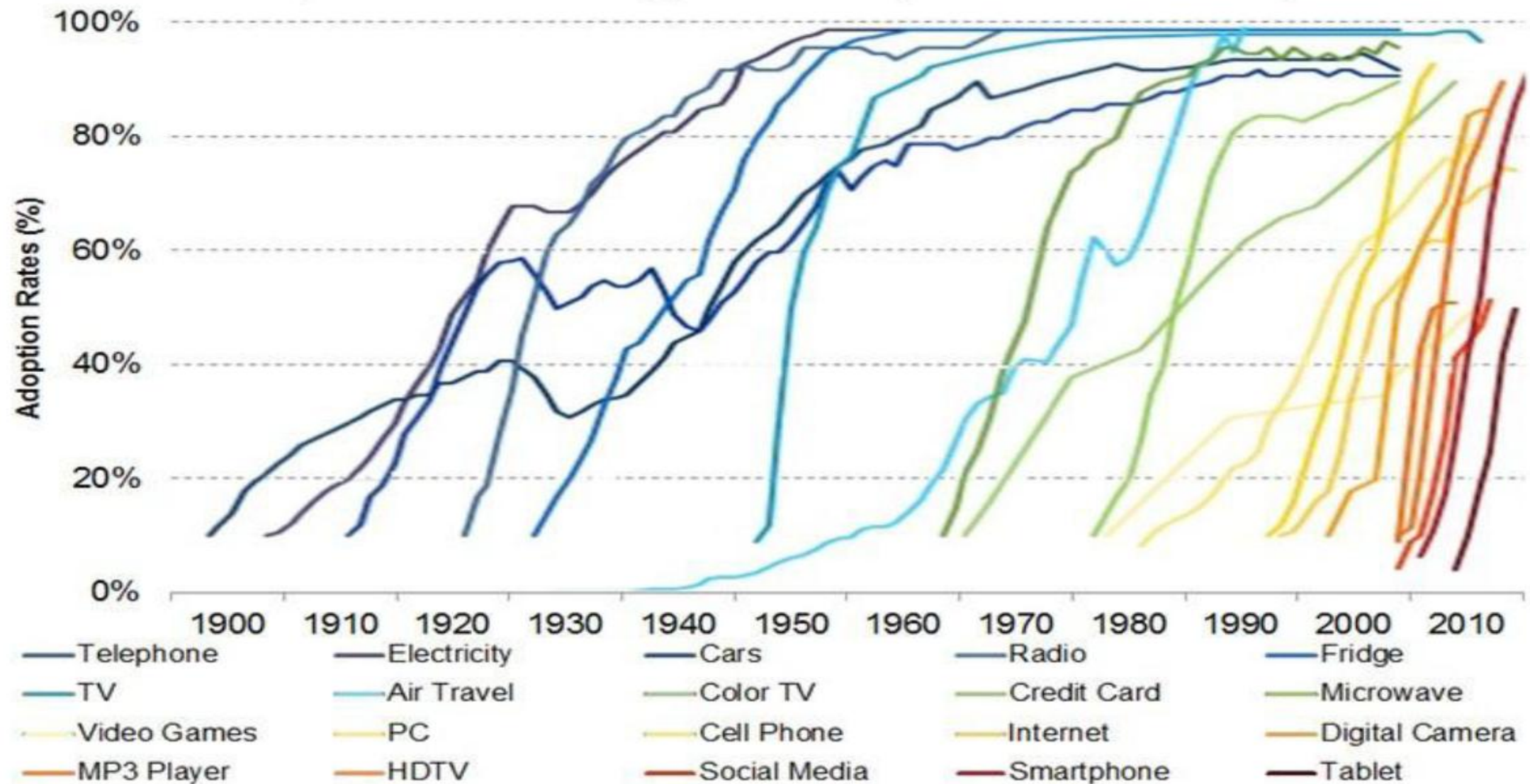


Agenda:

- Why AI
 - Impact of AI
 - AI Ethics
 - AI Ethics framework
 - Long term Social Impact of AI
- 

Why AI ?

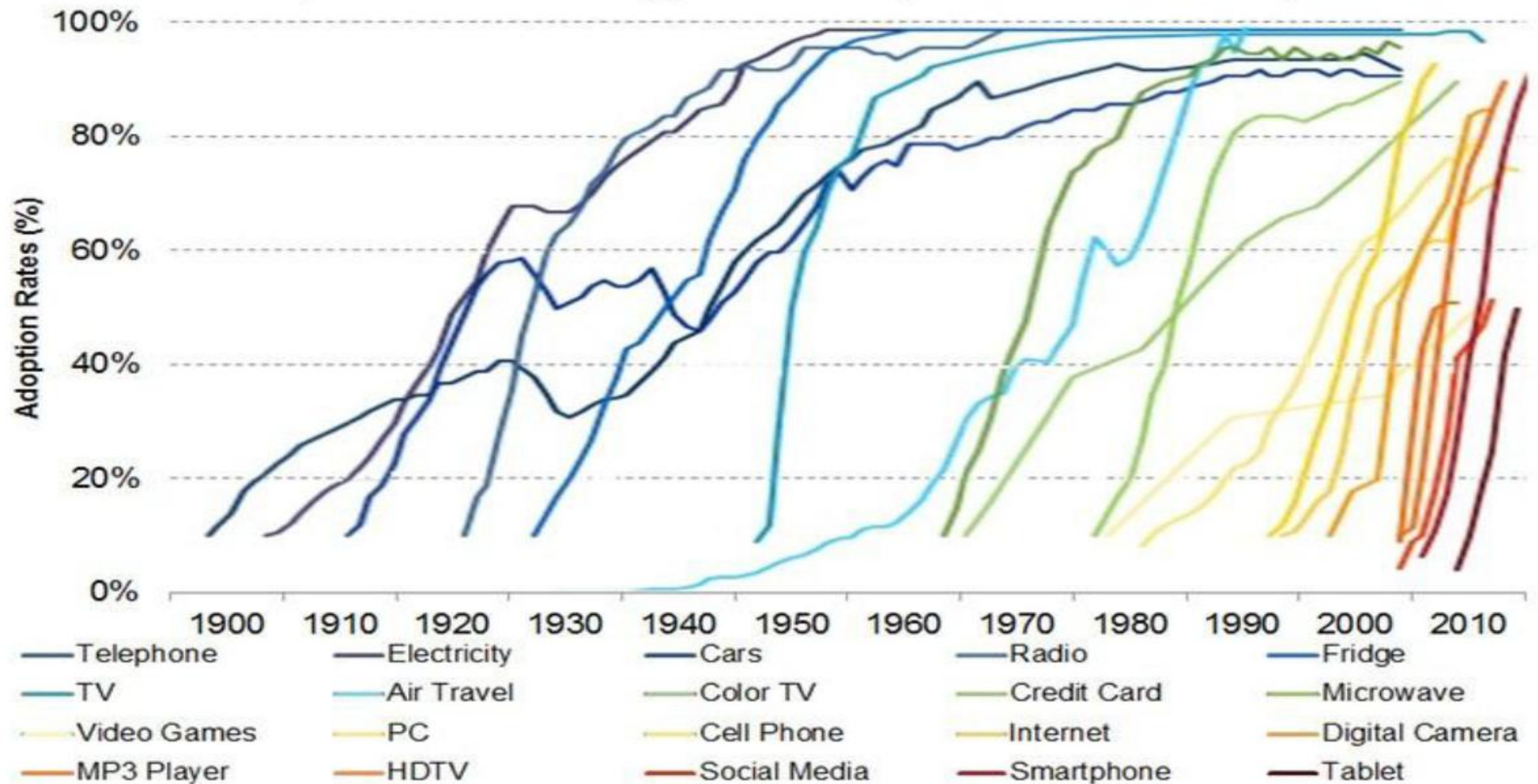
Industrial & consumer trends for last ~125 years



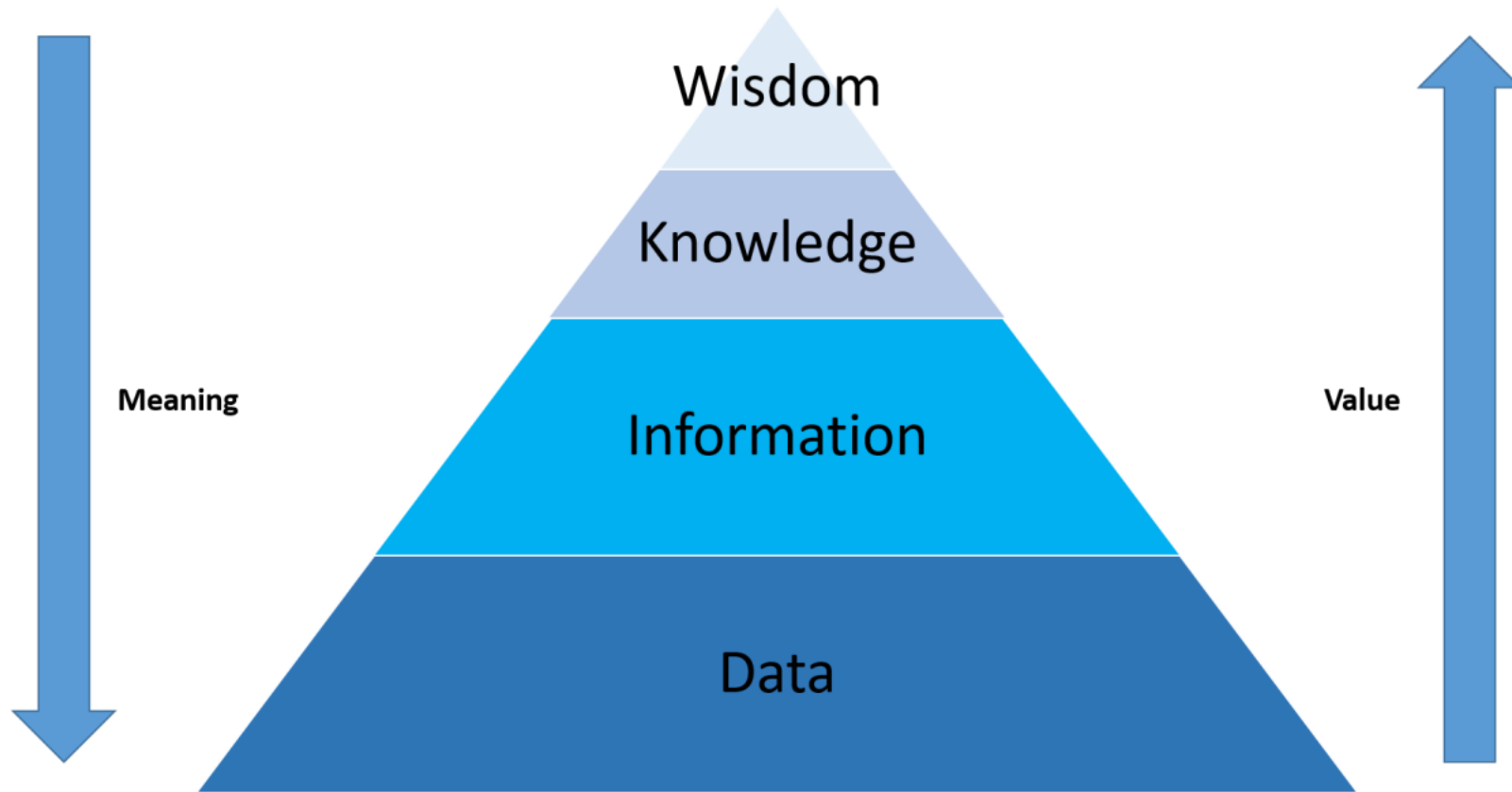
Mechanical + Electrical

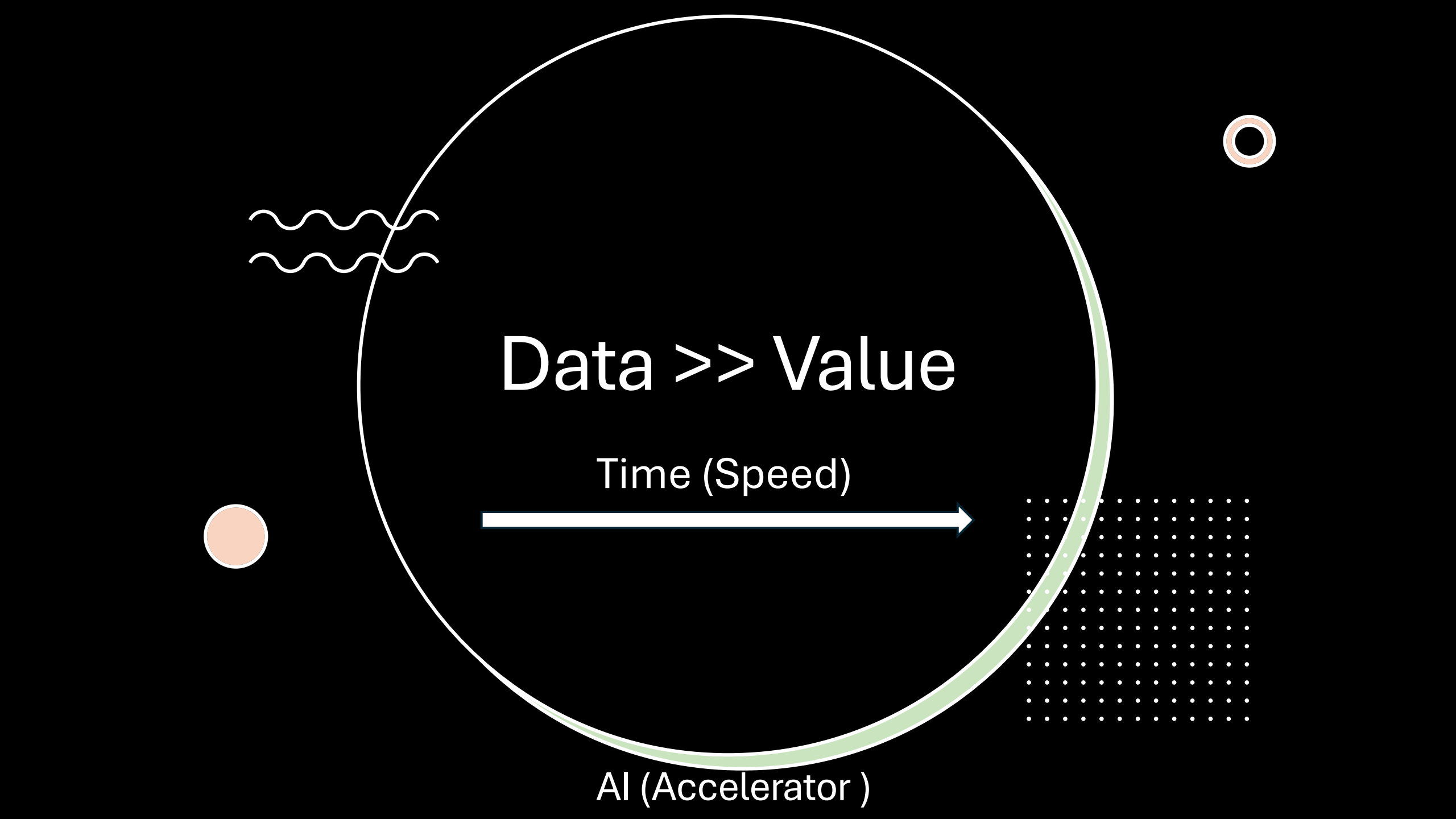
Electronic

Data



DIKW Pyramid of Data (Science)



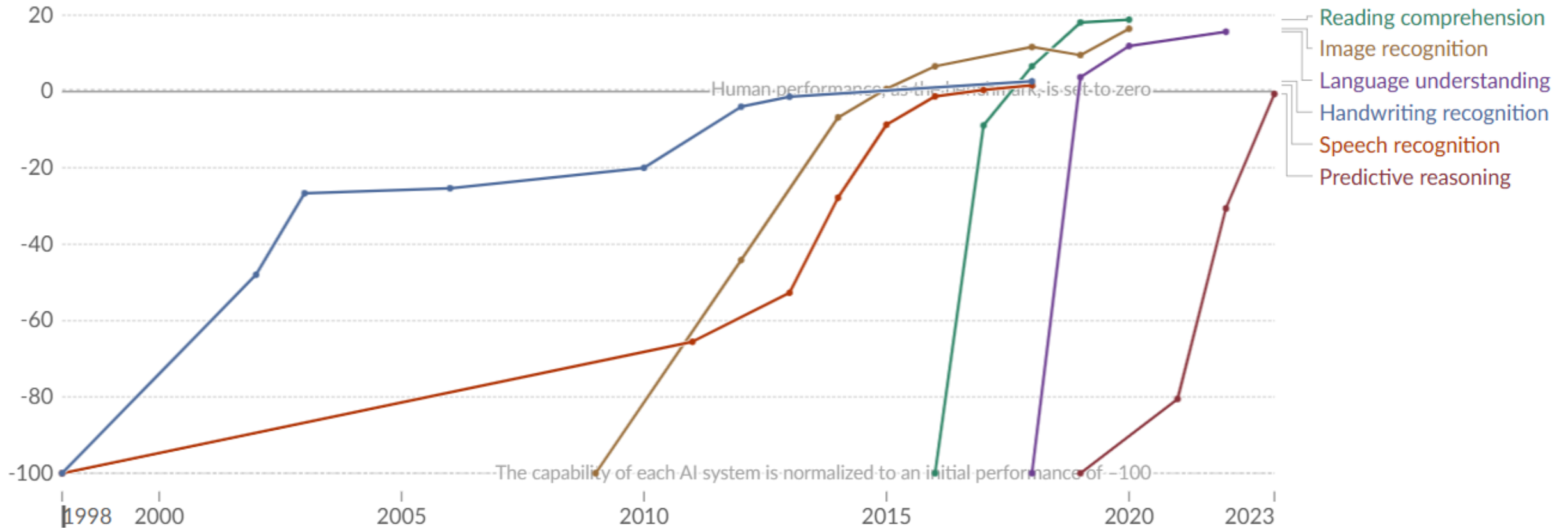


Data >> Value

Time (Speed)

AI (Accelerator)

Test scores of AI systems on various capabilities relative to human performance



AI Ethics

VOUS NE CONNAISSEZ PAS
CET HOMME. POURTANT,
IL A CHANGÉ NOS VIES.

BENEDICT
CUMBERBATCH
KEIRA
KNIGHTLEY
**IMITATION
GAME**
UN FILM DE
MORTEN TYLDUM

BLACK WAVE PICTURES'S version of the story of the brilliant mathematician who cracked the German Enigma code during World War II. Screenplay by the author of "The Imitation Game" and "The Turing Test" by the author of "The Turing Test" and "The Turing Test".

CASTING BY SAMANTHA SHOOTING BY JAMES W. HARRIS MUSIC BY JAMES NEWTON HOWARD COSTUME DESIGNER JAMES W. HARRIS EDITOR JAMES W. HARRIS PRODUCTION DESIGNER JAMES W. HARRIS EXECUTIVE PRODUCERS JAMES W. HARRIS PRODUCED BY JAMES W. HARRIS

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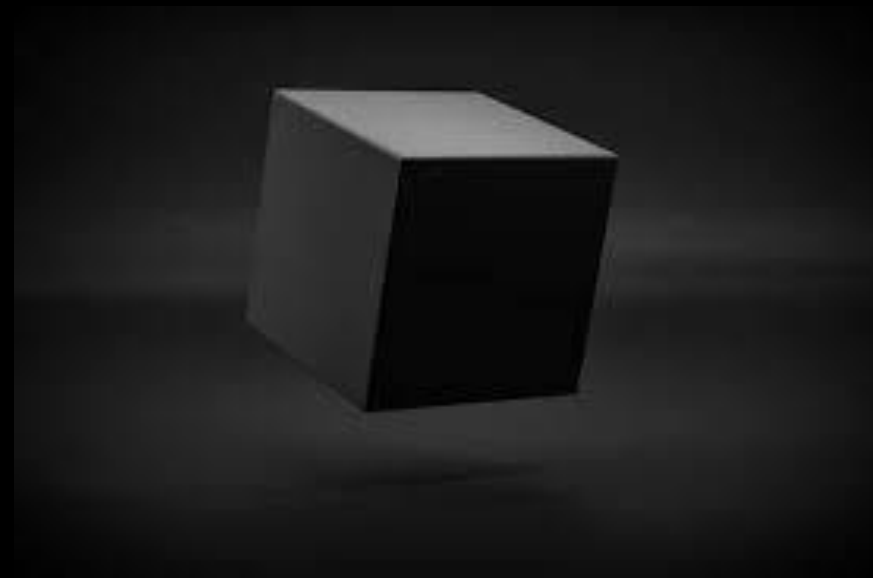
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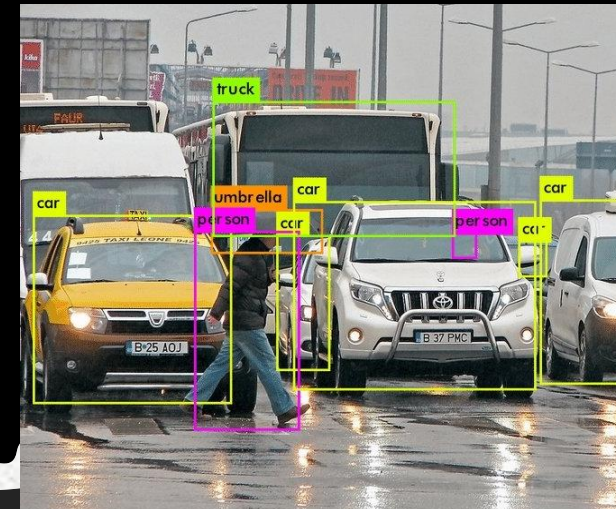
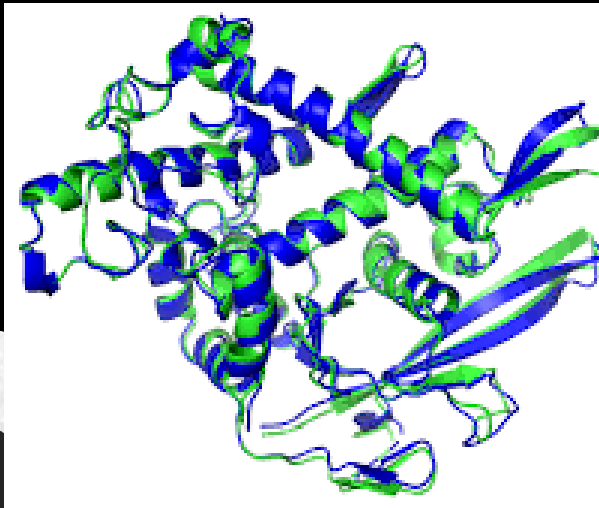
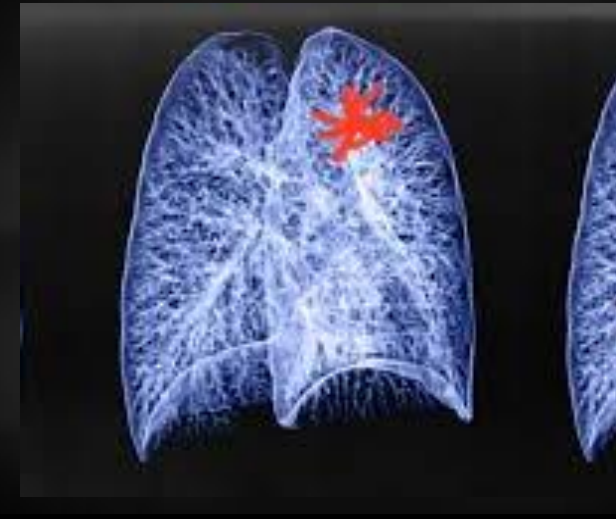
Are we at Wright
Brothers (1904)
Moment with AI ?



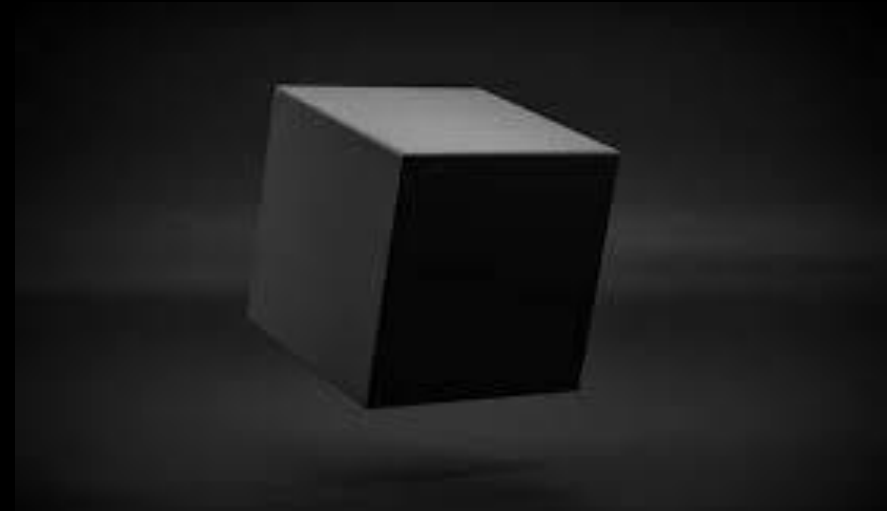
Missing the fundamentals
and can't explain current
AI capabilities work



AI Success Stories ..



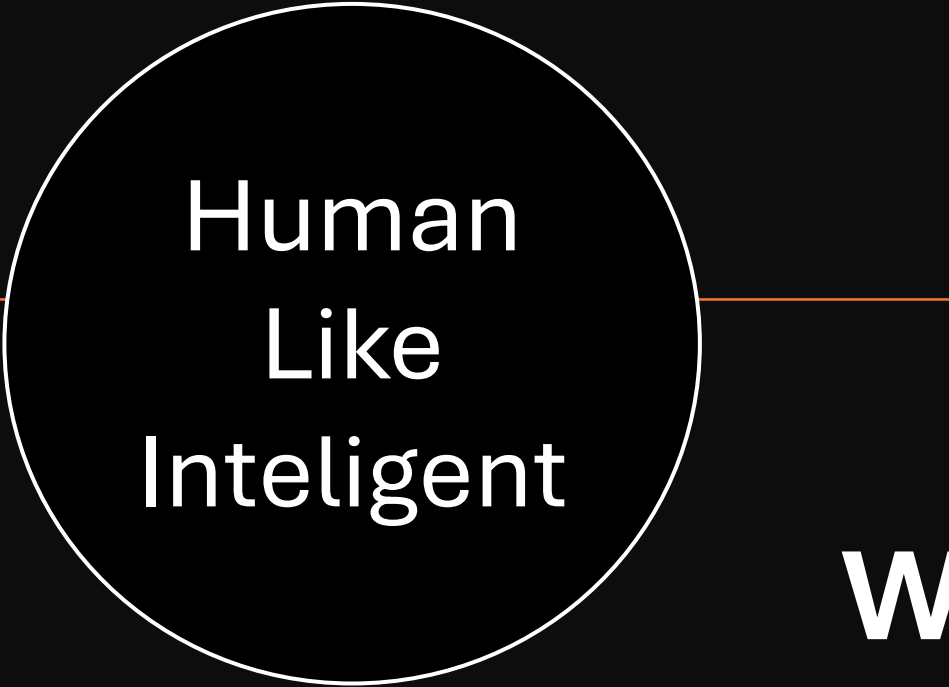
AI failed stories..





Machines
work with
human

Human
Like
Intelligent



Human
Like
Intelligent

**What are AI ethics ?
who sets them?**

AI ethics focus areas



Fairness & Bias
Mitigation



Transparency &
Explainability



Accountability &
Responsibility



Privacy & Data
Protection



Safety & Security



Sustainability

AI Regularization



In general, all industry and consumer protection & technology regulations are applicable for AI as well.

- EU's General Data Protection Regulation (GDPR) – 2018
- California Consumer Privacy Act (CCPA)
- Sri Lanka - Personal Data Protection Act (PDPA), No. 9 (2022)

AI Specific Regulations:

- European Union's Artificial Intelligence Act (EU AI Act)
- Generative AI Accountability Act - SB 896 (California)

AI Guidelines/Framework (Not regulations)

- NIST AI 600-1, Artificial Intelligence Risk Management Framework



Fairness & Bias Mitigation



Data, Algorithm, Design Deployment, Maintenance

- E.g. Facial recognition AI trained on Western datasets underperforms for darker skin tones.
- Mitigation strategies: Balanced datasets, fairness constraints in ML models.

Protected Classes

- Race,
- Color,
- Religion,
- Sex (including pregnancy, sexual orientation, or gender identity),
- National Origin
- family medical history

Other biased, but not protected classes.

- Culture
- Social-economic Status
- Income, employment status

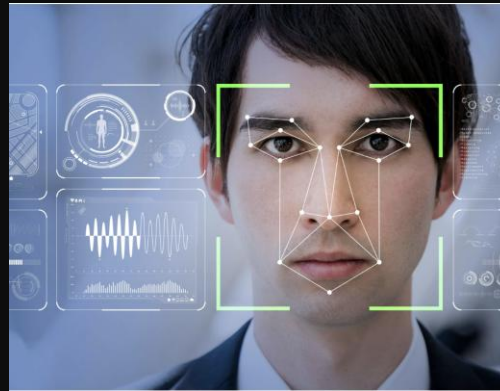


Transparency & Explainability

- Transparent on purpose, design, development, and deployment processes.
- Explain and document everything (if possible)
- Continues user feedback.
- 3rd party audits, work with regulators etc.
- User techniques like SHAP, LIME, interpretable model



Privacy & Data Protection



Issues:

- Deepfake fraud
- Profiling/Impersonation
- Unethical People ranking

Regulations:

- EU's GDPR
- (CCPA)
- Sri Lanka - Personal Data Protection Act (PDPA), No. 9 (2022)

Solutions

- Data Encryptions
- MFAS



Safety & Security

- AI systems controlling critical infrastructure must be fail-safe.
- Risks of AI in cybersecurity: Deepfake fraud, automated phishing attacks.
- Securing AI systems against adversarial attacks.



Accountability & Responsibility



Who is responsible for Boeing 737 MAX crashes and the role of automation failures ?

- Who is responsible for AI-caused harm? (Developers, businesses, governments?)
- Need for AI governance frameworks in organizations.



Sustainability



- high Energy Consumption
- Carbon footprint/impact on global warming
- High Resource Utilization



Social Impact of AI

- Human Longevity
 - Personalized medication & medication
- Workforce Transformation & Job Displacement
 - Data is the new asset
- **AI taxation/Universal Basic Income (UBI)**
- **AI-augmented human**



How Should We Prepare?

- Governments need robust AI policies to address long-term risks.
- Individuals must develop AI literacy to adapt to the evolving AI landscape.
- Companies must prioritize responsible AI to mitigate social harms.
- Academia and researchers must focus on AI alignment to ensure ethical AI systems.